Progress and Preservation

Concepts of Programming Languages

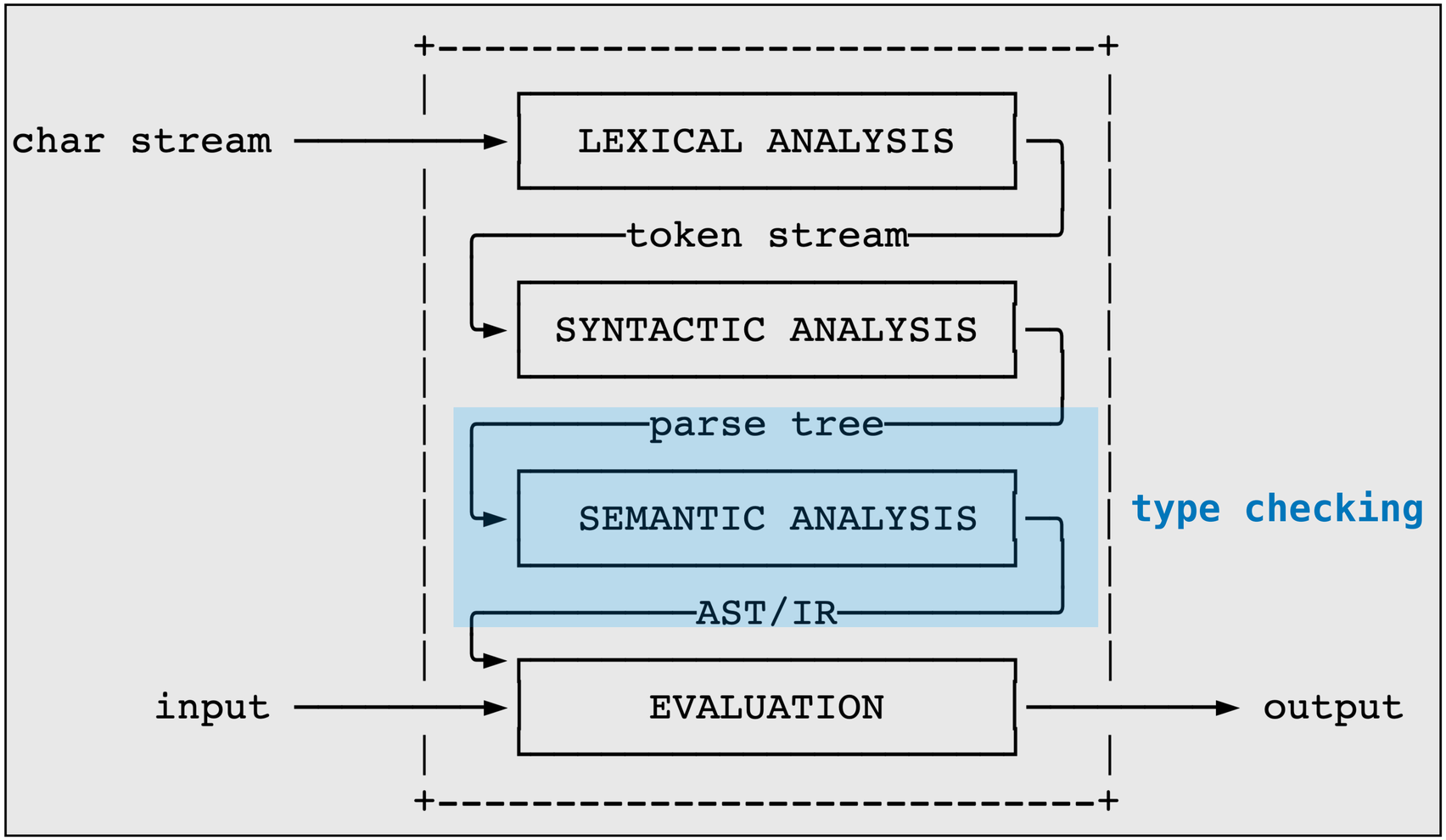
CAS CS 320

Outline

- Discuss **induction** over derivations
- Show that STLC satisfies **progress** and **preservation**

Recap

Recall: The Picture



Recall: Type Checking/Inference

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type_check : expr -> ty -> bool
type_of : expr -> ty option
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Type checking: determining whether an expression can be typed

Type inference: *synthesizing* a type for an expression

Theoretically, these two problems can be very different. *For STLC, they are both easy*

Recall: Syntax (STLC)

$$e ::= \bullet \mid x \mid \lambda x^{\tau}.\, e \mid ee$$

$$\tau ::= \top \mid \tau \to \tau$$

$$x ::= variables$$

The syntax is the same as that of the lambda calculus except:

- » we include a unit expression
- » we have types, which annotate arguments

Recall: Typing (STLC)

$$\frac{}{\Gamma \vdash \bullet : \top}\ \texttt{unit}$$

$$\frac{\Gamma, x : \tau \vdash e : \tau'}{\Gamma \vdash \lambda x^{\tau}.\, e : \tau \to \tau'}\ \texttt{abstraction}$$

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \texttt{variable}$$

$$\frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'}\ \texttt{application}$$

These rules enforce that a function can only be applied if we *know* that it's a function

Recall: Semantics (STLC)

$$\frac{e_1 \longrightarrow e_1'}{e_1 e_2 \longrightarrow e_1' e_2}\ \texttt{leftEval}$$

$$\frac{}{(\lambda x\,.\,e)e' \longrightarrow [e'/x]e}\ \texttt{beta}$$

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This is part of the point. Type-checking only determines *whether* we go on to evaluate the program

It doesn't determine *how* we evaluate the program

Practice Problem

$$(\lambda x^{(\top \to \top) \to \top} . x(\lambda z^{\top} . x(wz)))y$$

Determine the smallest context such that the above expression is typeable (also give its type)

Answer

$$(\lambda x^{(\top\to\top)\to\top}.\,x(\lambda z^{\top}.\,x(wz)))y$$

Induction over Derivations

The Key Idea

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$$\dfrac{\dfrac{}{\{\} \vdash \texttt{2} : \texttt{int}}\,(\text{intLit}) \qquad \dfrac{\dfrac{}{\{\texttt{y} : \texttt{int}\} \vdash \texttt{y} : \texttt{int}}\,(\text{var}) \qquad \dfrac{}{\{\texttt{y} : \texttt{int}\} \vdash \texttt{y} : \texttt{int}}\,(\text{var})}{\{\texttt{y} : \texttt{int}\} \vdash \texttt{y} + \texttt{y} : \texttt{int}}\,(\text{intAdd})}{\{\} \vdash \texttt{let y = 2 in y} + \texttt{y} : \texttt{int}}\,(\text{let})$$

Derivations are *trees*

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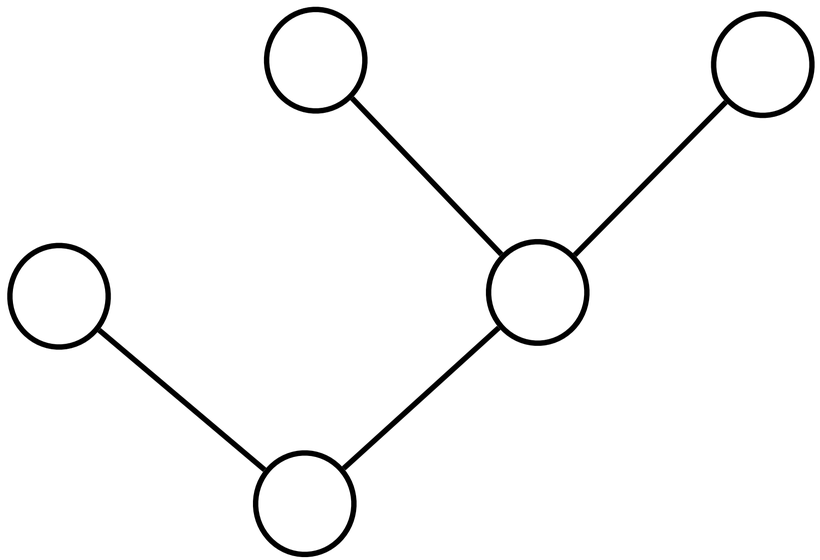
Derivations are *trees*

We can prove things about trees using induction



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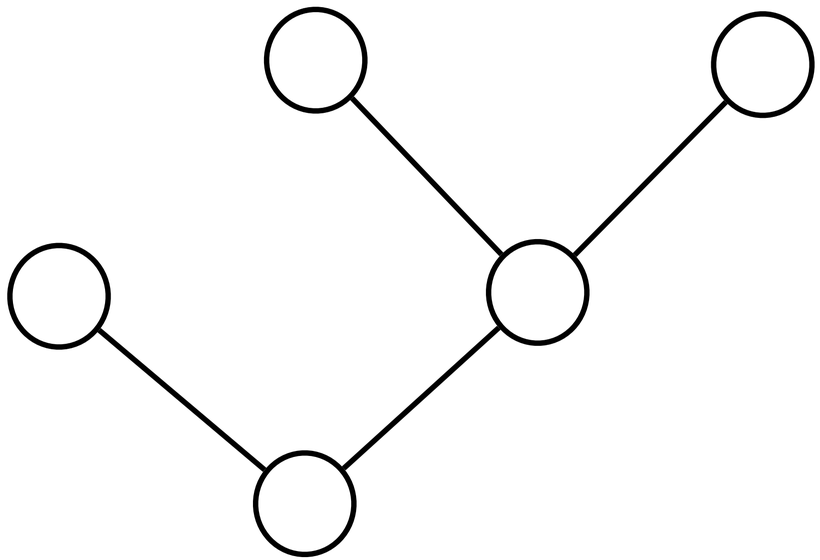

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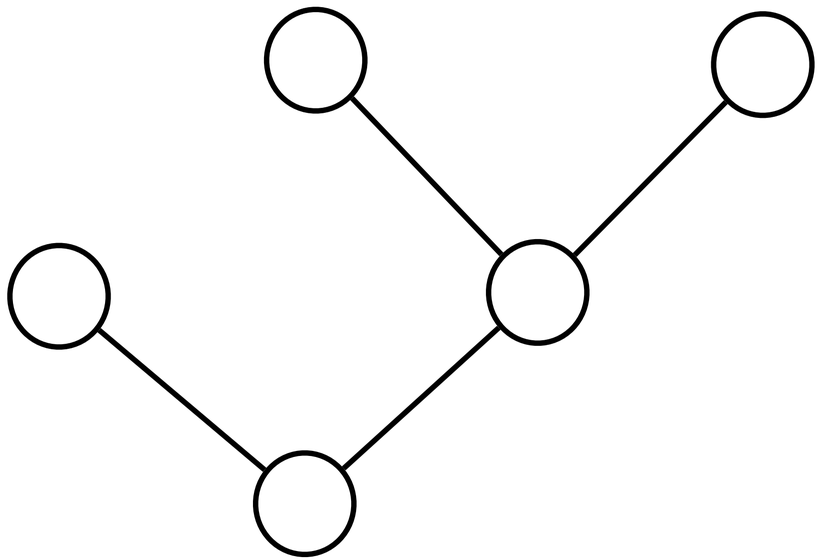
Important: Every derivable judgment corresponds to a derivation



Warm-up: Binary Trees

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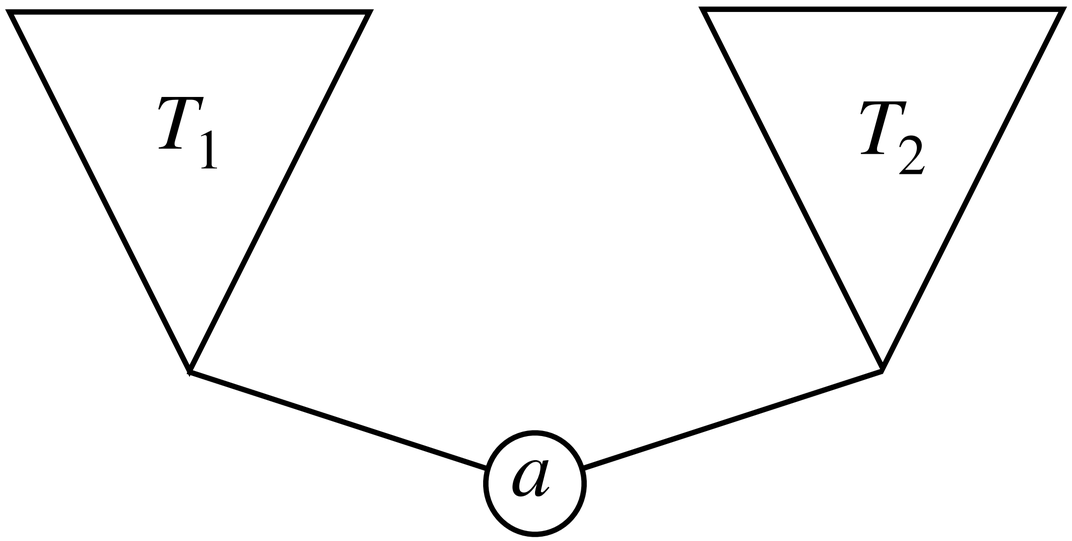
Theorem. $\text{size}(T) \leq 2^{\text{height}(T)} - 1$ for any tree T

Proof. By induction on the *structure of trees*

Base Case: Empty

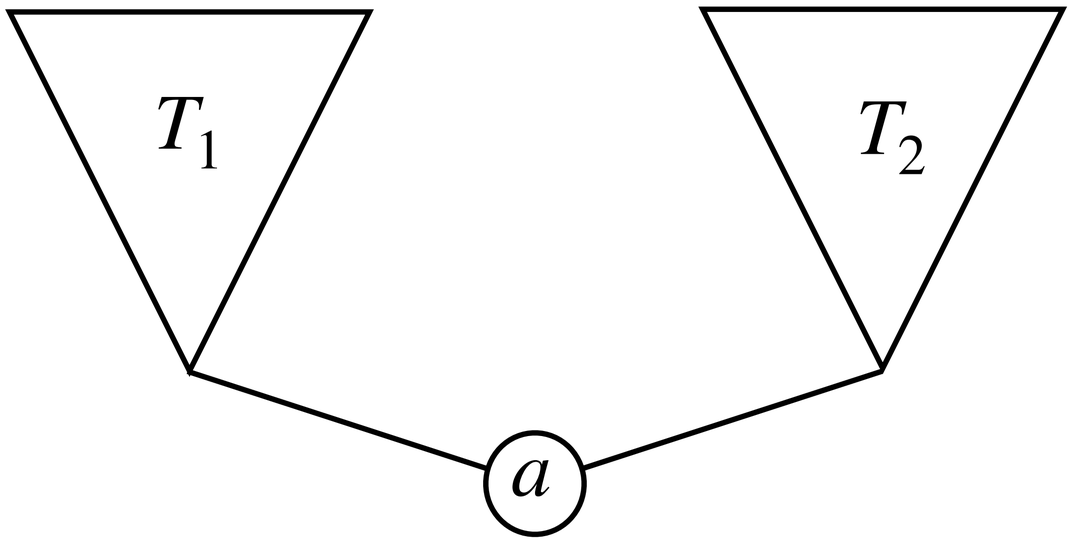
Inductive Hypothesis



If T is of the form `Node` $(T_1,\ a,\ T_2)$ then $\text{size}(T_1) \leq 2^{\text{height}(T_1)} - 1$ and $\text{size}(T_2) \leq 2^{\text{height}(T_2)} - 1$

That is, we get to assume that what we want holds of our subtrees

Inductive Step: Nodes



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An expression e is **well-scoped** with respect to a context Γ if $x \in FV(e)$ implies x appears in Γ

Theorem. If e is well-typed in Γ, then e is well-scoped

Proof. By induction on *derivations*

Base Case: Axioms

$$\frac{}{\Gamma \vdash \bullet : \top}\ \texttt{unit}$$

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \texttt{variable}$$

We need to show that expressions typed using just axioms satisfy well-scopedness

Inductive Hypothesis

$$\frac{\begin{array}{c}\vdots\\ \mathscr{D}_1\\ \Gamma_1 \vdash e_1 : \tau_1\end{array} \quad \begin{array}{c}\vdots\\ \mathscr{D}_2\\ \Gamma_2 \vdash e_2 : \tau_2\end{array} \quad \ldots \quad \begin{array}{c}\vdots\\ \mathscr{D}_k\\ \Gamma_k \vdash e_k : \tau_k\end{array}}{\Gamma \vdash e : \tau}$$

If $e_1, \ldots, e_k$ *are well-scoped (because they are typeable in the each of their contexts)*

Inductive Step 1: Application

$$\dfrac{\begin{array}{c}\vdots \\ \mathscr{D}_1 \\ \Gamma \vdash e_1 : \tau \to \tau'\end{array} \qquad \begin{array}{c}\vdots \\ \mathscr{D}_2 \\ \Gamma \vdash e_2 : \tau\end{array}}{\Gamma \vdash e_1 e_2 : \tau'}\ \texttt{application}$$

What if the last rule I applied was application?

Inductive Step 2: Abstraction

$$\dfrac{\begin{array}{c}\vdots \\ \mathscr{D} \\ \Gamma, x : \tau \vdash e : \tau'\end{array}}{\Gamma \vdash \lambda x^{\tau}.\, e : \tau \to \tau'}\ \texttt{abstraction}$$

What if the last rule I applied was abstraction?

Progress and Preservation

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With small-step semantics, we can give a finer-grained analysis:

> **Thm.** If $\cdot \vdash e : \tau$, then
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> » *(progress)* either e is a value or there is an e' such that $e \longrightarrow e'$
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> » *(preservation)* If $\cdot \vdash e : \tau$ and $e \longrightarrow e'$ then $\cdot \vdash e' : \tau$

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These results are *fundamental*. They tell us that our PL is well-behaved (it's a "good" PL)

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With small-step semantics, we can give a finer-grained analysis:

goal for today

> **Thm.** If $\cdot \vdash e : \tau$, then
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These results are *fundamental*. They tell us that our PL is well-behaved (it's a "good" PL)

Disclaimer: We're gonna hand-wave liberally

Recall: STLC

Syntax

$$e ::= \bullet \mid x \mid \lambda x^{\tau}.\, e \mid ee$$

$$\tau ::= \top \mid \tau \to \tau$$

$$x ::= \mathit{variables}$$

Typing

$$\frac{}{\Gamma \vdash \bullet : \top}\ \texttt{unit}$$

$$\frac{\Gamma, x : \tau \vdash e : \tau'}{\Gamma \vdash \lambda x^{\tau}.\, e : \tau \to \tau'}\ \texttt{abstraction}$$

$$\frac{(x : \tau) \in \Gamma}{\Gamma \vdash x : \tau}\ \texttt{variable}$$

$$\frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'}\ \texttt{application}$$

Semantics

$$\frac{e_1 \longrightarrow e_1'}{e_1 e_2 \longrightarrow e_1' e_2}\ \texttt{leftEval}$$

$$\frac{}{(\lambda x.\, e)e' \longrightarrow [e'/x]e}\ \texttt{beta}$$

Progress (STLC)

> **Theorem.** If e is well-typed then e is a value or there is an expression e' such that $e \longrightarrow e'$

Proof. By induction over *derivations*

Base Case: Axioms

$$\frac{}{\cdot \vdash \bullet : \top}\ \texttt{unit}$$

$$\frac{(x : \tau) \in \varnothing}{\cdot \vdash x : \tau}\ \texttt{variable}$$

We need to show that expressions typed using just axioms yield non-stuck terms

Inductive Step 1: Application

$$\frac{\cdot \vdash e_1 : \tau \to \tau' \qquad \cdot \vdash e_2 : \tau}{\cdot \vdash e_1 e_2 : \tau'} \ \texttt{application}$$

*What do we know given that e_1 is either a **value** or **reducible**?*

Inductive Step 2: Abstraction

$$\frac{\{x : \tau\} \vdash e : \tau'}{\cdot \vdash \lambda x^{\tau}.\, e : \tau \to \tau'}\ \texttt{abstraction}$$

Our expression already a value if the last rule we applied was abstraction!

Preservation (STLC)

Theorem. If $\Gamma \vdash e : \tau$ and $e \longrightarrow e'$ then $\Gamma \vdash e' : \tau$

Proof. By induction over *derivations*

This one is much tricker...

Base Case: Axioms

$$\frac{}{\Gamma \vdash \bullet : \top}\ \texttt{unit}$$

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Expressions typed using just axioms cannot be reduced (nothing to do here)

Inductive Step 1: Abstraction

$$\frac{\Gamma, x : \tau \vdash e : \tau'}{\Gamma \vdash \lambda x^{\tau} . e : \tau \to \tau'} \text{ abstraction}$$

Expressions derived using abstraction as the last rule is already a value (nothing to do here)

Inductive Step 2: Application

$$\frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'} \text{ application}$$

This is where the work comes in...

The trick: We do induction (inside our current induction) on the structure of *semantic* derivations!

What possible ways can $e_1 e_2$ *be reduced?*

Inductive Step 2.1: leftEval

$$\frac{e_1 \longrightarrow e_1'}{e_1 e_2 \longrightarrow e_1' e_2}\ \texttt{leftEval} \qquad \frac{\Gamma \vdash e_1 : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash e_1 e_2 : \tau'}\ \texttt{application}$$

What if our last rule was an application ***and*** $e_1 e_2$ *is reducible by leftEval?*

Inductive Step 2.1: leftEval

$$\frac{}{(\lambda x\,.\,e)e_2 \longrightarrow [e_2/x]e}\ \texttt{beta} \qquad \frac{\Gamma \vdash (\lambda x\,.\,e) : \tau \to \tau' \qquad \Gamma \vdash e_2 : \tau}{\Gamma \vdash (\lambda x\,.\,e)e_2 : \tau'}\ \texttt{application}$$

What if our last rule was an application ***and*** $e_1 e_2$ *is reducible by beta?*

Substitution Lemma

Lemma. If $\Gamma \vdash e_2 : \tau_2$ and $\Gamma, x : \tau_2 \vdash e : \tau$ then

$$\Gamma \vdash [e_2/x]e : \tau$$

That is, if e is well-typed in a context with (x, τ) then we can substitute x with anything of type τ and it's still the same type

(we can prove this by, you guessed it, induction on derivations)

The Point

```
let rec eval env e =
  match e with
  | Var x -> Env.find x env
  ...
```

Progress and preservation tell us that **terms never get stuck during evaluation**

This is huge. I can't emphasize this enough

Our type system ensures we only evaluate programs that make sense!

Summary

Progress and **preservation** are fundamental features of good programming languages

We can prove things about well-typed expressions by performing **induction** over derivations